

RESEARCH AGENDA 2026-2030

*Biosphere Reserve of the yApayaos
(BRyA)*

INTRODUCTION

Background

The Biosphere Reserve of the yApayaos (BRyA) was designated under UNESCO's Man and the Biosphere (MAB) Programme on July 5, 2024, marking the province as a globally recognized site for integrating conservation, sustainable development, and cultural resilience. This recognition underscores the need for management strategies and policies informed by robust, interdisciplinary, and Indigenous knowledge-grounded research.

As a developing province, formally established in February 14, 1995 following its separation from Kalinga under Republic Act 7878, Apayao faces challenges in building a research-based foundation that supports the sustainable resource use and development of the BRyA as part of the global network of the MAB programme.

Objectives of the Research Agenda

In accordance with the technical guidelines of the UNESCO Man and Biosphere Programme, Biosphere Reserves must satisfy three core functions: conservation, development, and logistic support. To fulfill these functions, policymaking, management, and decisionmaking in the province must be data-based, inclusive, and grounded in scientific evidence. Given Apayao's unique ecological and cultural landscape, research must be holistic not only to address environmental priorities but also to integrate and recognize traditional knowledge as central to sustainable development within the province.

The BRyA Research Agenda (2026–2030) therefore seeks to provide a strategic framework that will guide, coordinate, and strengthen research initiatives across disciplines and institutions. It aims to ensure that studies conducted within Apayao are aligned with local priorities, contribute to the MAB Programme's global objectives, and serve as the basis for evidence-informed planning, capacity building, and policy formulation in the province. To achieve this goal, the research agenda outlines the following objectives:

1. To identify gaps and priority research areas for the Biosphere Reserve of the yApayaos;
2. To define institutional arrangements, funding prioritization, and monitoring and evaluation mechanisms to support research initiatives; and
3. To establish a framework for partnerships, collaboration, and knowledge-sharing among key stakeholders of the BRyA.

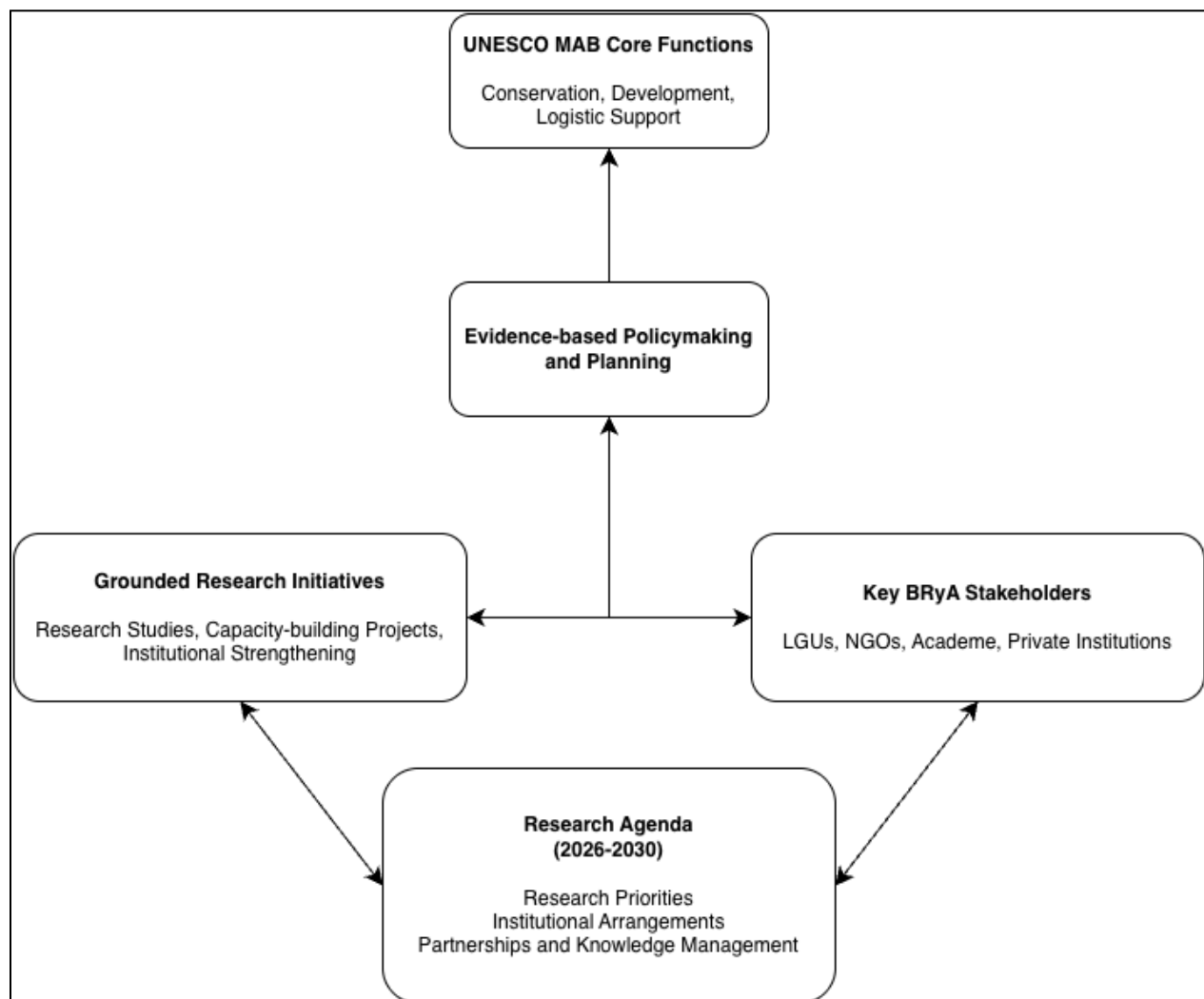
RESEARCH AGENDA FRAMEWORKS

Conceptual Framework

The BRyA Research Agenda seeks to illustrate the key processes, outputs, and feedback mechanisms that will enable the province to achieve its overarching goal: to support and operationalize the three core functions of a biosphere reserve (conservation, development, and logistic support) under UNESCO's Man and the Biosphere (MAB) Programme.

Figure N.

BRyA Research Agenda Conceptual Framework



RESEARCH AGENDA

The BRyA Research Agenda for 2025–2030 identifies four overarching thematic priorities that will guide future research and knowledge-generation efforts within the province: (1) Biodiversity Research and Conservation, (2) Land Use and Natural Resource Management, (3) Climate Change and Ecosystem Resilience, and (4) Sociocultural and Indigenous Knowledge Systems and Practices (IKSPs).

Each thematic priority outlines knowledge gaps, priority research topics, and desired outputs. These were updated and validated during the 2025 BRyA Research Summit, held on 10–11 October 2025, where representatives from various academic research institutions, government agencies, and non government organizations participated in engaging discussions to collectively determine the province’s research needs and opportunities for the next planning cycle.

Research topics and corresponding research questions were also formulated for each thematic research priority to guide researchers, HEIs, and agencies in designing and developing studies. This ensures research outputs that guide the development of programs, projects and activities that bridge the knowledge and policy gaps in the short- and long-term conservation and sustainable development goals of the province.

Land Use and Natural Resource Management

Central to the discussion of the effective management of the Biosphere Reserve of the yApayaos (BRyA) is land use and natural resource management. Policymaking and decision-making mechanisms for zoning, issuance of tenurial instruments, formulation of land use plans, and ecosystem valuation among many others is critical to ensuring that the BRyA supports sustainable development within the province. The 2025 BRyA Research Summit identified gaps in the management of zoning and land tenure conflicts, insufficient documentation, and low community awareness.

This theme aims to produce scientific and community-grounded knowledge that supports coherent land-use decisions and long-term ecosystem management.

Fundamental Gaps

1. Zoning conflicts and multi-issuance of tenurial instruments
2. Lack of harmonized boundaries between provincial-level zoning and municipal zoning (i.e. BRyA Core Zones vs municipal surveys)
3. Insufficient documentation of land disputes
4. No institutionalized land-use suitability assessments
5. Limited quantitative data on ecosystem services
6. Low community awareness and CEPA monitoring

Priority Research Areas

1. Zoning Conflicts and Multi-Issuance of Tenurial Instruments
 - a. Assessment of overlapping tenurial instruments (e.g., CADT, CBFMA, titles)
 - b. Mapping and analysis of zoning and tenurial inconsistencies across agencies
 - c. Socioeconomic impacts of land tenure conflicts
 - d. Institutional pathways leading to multi-issuance
 - e. Development of conflict resolution frameworks relevant to BRyA
2. Harmonization of BRyA Zonation and Municipal Land-Use Plans
 - a. Comparative spatial analysis of BRyA core/buffer/transition zones and municipal CLUPs
 - b. Field validation of boundary discrepancies
 - c. Impacts of zoning mismatches on enforcement and development
 - d. LGU capacity assessment for integrating BRyA boundaries

- e. Development of a unified BRyA–LGU zoning map
- 3. Documentation and Analysis of Land Disputes
 - a. Inventory and typology of land disputes (resource use, boundaries, tenurial claims)
 - b. Socio-cultural and historical drivers of land conflicts
 - c. Mapping dispute hotspots
 - d. Evaluation of dispute resolution mechanisms
 - e. Conflict sensitivity and community-based mediation strategies
- 4. Institutionalized Land-Use Suitability Assessments
 - a. Land capability and suitability assessments for agriculture, habitation, tourism, and conservation
 - b. Soil, slope, hazard, and watershed integrity analysis
 - c. Suitability indices and decision-support tools
 - d. Feasibility of sustainable land-use alternatives
 - e. Integration of ecosystem-based approaches into planning processes
- 5. Quantifying Ecosystem Services
 - a. Valuation of ecosystem services (water supply, carbon storage, biodiversity)
 - b. Identification of service hotspots and priority protection areas
 - c. Community perception studies on ecosystem service benefits
 - d. Natural capital accounting approaches for provincial planning
 - e. Integration of ecosystem service metrics into CLUPs and investment plans
- 6. Community Awareness and CEPA Monitoring
 - a. Community awareness assessments on zoning, environmental laws, and BRyA rules
 - b. Evaluation of CEPA strategies and barriers to public understanding
 - c. Development of localized IECs and educational tools
 - d. Behavioral insights research for improved compliance
 - e. Youth and school-based CEPA program design

Expected Outputs

By 2031, research initiatives developed under this theme and with the research agenda as the guiding framework are expected to produce:

1. Harmonized BRyA–LGU Zoning Map with validated boundaries
2. Geospatial inventory of overlapping tenurial instruments and conflict-prone zones

3. Provincial Land Dispute Registry and technical report on dispute drivers and patterns
4. Guidelines for issuing, harmonizing, and monitoring tenurial instruments across agencies
5. Land Suitability Maps (agricultural, settlement, conservation, hazard-sensitive areas)
6. Ecosystem Services Valuation Report for key watersheds and landscapes
7. Decision-support tools for integrating ecosystem services and suitability into planning
8. CEPA Monitoring Framework and community awareness baseline reports
9. IEC toolkits for barangays, schools, and IP communities on land-use rules and zoning
10. Policy briefs and recommendations for land-use harmonization, dispute resolution, and sustainable development

IMPLEMENTATION AND GOVERNANCE

Institutional Arrangements

The lead implementing office for the research agenda will be the **BRyA Management Division** under the Provincial Government of Apayao. The advisory board for implementing the research agenda will be the **Biosphere Reserve Research Sub-committee** under the Committee on Logistics of the BRyA Council. The BRyA Management Division will have the following functions:

1. Reviewing and endorsing research proposals to be conducted within the BRyA
2. Managing research permits, FPIC pathways, and ethical compliance
3. Coordinating with LGUs, ENRO, NCIP, DENR, HEIs, and IP organizations
4. Coordination with NCIP, DOST, DENR for permit clearances

Assisting with the implementation of research initiatives within the province will be the respective Municipal Environment and Resource Office (MENRO) where the research will be situated. The MENROs will have the following functions:

1. Local data gathering and validation
2. Facilitating ground access, safety support, and community entry
3. Monitoring research activities within their jurisdiction
4. Integrating findings into CLUPs, DRRM plans, and municipal PPAs

Funding Prioritization

As funding from the provincial government office is limited, funding principles and criteria have been developed to guide funding prioritization. These principles and criteria shall be implemented by the relevant management bodies to evaluate grant proposals for research initiatives within the province. A scoring matrix may be developed based on these said principles and criteria for approval. It is also highly recommended that standard operating procedures be created to guide the grant and subsidy proposal submission process.

Funding Principles/Criteria

1. Alignment with BRyA Research Agenda (core requirement)
2. Community-benefit and relevance to local governance
3. Urgency and risk reduction (e.g., climate hazards, biodiversity threats)
4. Inclusion of local researchers, youth, and IPs
5. Sustainability and long-term value of outputs

Priority Funding Categories

Category	Description
Foundational Research	Baseline biodiversity surveys, land-use mapping, climate hazard mapping
Policy-Enabling Research	Studies that directly inform LGU planning, protected area management, NCIP coordination
High-Impact Innovation Research	Renewable energy pilots, ecosystem modeling, digital platforms
Community-Led or IP-Led Research	IKSP documentation, Lapat studies, socio-cultural mapping
Youth and Local Capacity Strengthening	Student-led projects, para-biologist programs, training workshops

Research Flowchart

The BRyA Management Division under the Provincial Government of Apayao will also be responsible for assisting researchers who are interested in pursuing research initiatives within the BRyA. As seen in the figure below, a research flowchart was also created as a guide for interested HEIs, organizations, or independent researchers to conduct research studies and initiatives. It is recommended that a monitoring and evaluation system be established to monitor and evaluate adherence to the processes indicated in the flowchart.

Figure N.

BRyA Research Flowchart

